

BORYSPIL, Ukraine

WMO: 333470

Lat: 50.3339N

Lon: 30.9492E

Elev: 122

StdP: 99.87

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-16.1	-22.2	0.5	-19.5	-18.9	0.7	-15.9	10.3	-3.6	9.4	-3.7	2.7	20	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.9	20.0	30.0	19.6	28.2	18.7	21.6	28.2	20.9	27.2	20.0	26.2	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	14.7	24.2	18.9	13.9	23.2	18.0	13.1	22.5	63.7	28.4	60.8	27.2	58.2	26.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.3	DB	-23.6	34.3	3.9	2.1	-26.4	35.9	-28.7	37.1	-30.9	38.3	-33.7	39.9	(4)
(5)			WB	-24.1	23.5	3.9	0.8	-26.9	24.0	-29.2	24.5	-31.4	25.0	-34.3	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-4.4	-3.3	1.7	9.3	15.2	18.5	20.6	19.5	14.1	7.7	2.4	-2.4	(6)
(7)		DBStd	10.00	5.87	6.08	4.63	4.15	4.21	3.41	2.96	3.35	3.86	4.40	4.91	5.59	(7)
(8)		HDD10.0	1864	446	372	259	62	7	0	0	8	97	230	384		(8)
(9)		HDD18.3	3871	704	606	517	272	114	39	12	26	134	328	477	642	(9)
(10)		CDD10.0	1250	0	0	1	41	168	257	328	294	132	27	2	0	(10)
(11)		CDD18.3	215	0	0	0	1	17	46	82	61	9	0	0	0	(11)
(12)		CDH23.3	2465	0	0	0	13	218	511	851	732	135	4	0	0	(12)
(13)		CDH26.7	776	0	0	0	2	46	142	286	269	30	0	0	0	(13)
(14)	Wind	WSAvg	3.7	4.0	4.1	4.2	3.9	3.6	3.5	3.2	3.1	3.3	3.4	3.9	3.9	(14)
(15)	Precipitation	PrecAvg	559	31	32	34	36	57	72	68	57	54	40	41	38	(15)
(16)		PrecMax	706	55	75	107	109	161	157	155	116	187	107	84	121	(16)
(17)		PrecMin	417	6	6	2	0	19	4	19	2	0	0	0	7	(17)
(18)		PrecStd	68	14	16	23	24	34	38	32	31	48	26	20	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	9.2	16.2	25.0	30.2	32.0	33.9	35.2	29.9	23.2	15.8	10.0	(19)
(20)			MCWB	5.2	6.5	9.2	15.2	18.4	19.9	20.8	20.5	17.8	14.9	12.4	8.8	(20)
(21)		2%	DB	5.1	6.9	13.2	21.8	27.9	30.0	31.8	32.1	26.9	20.1	12.8	7.2	(21)
(22)			MCWB	4.0	5.1	7.7	12.6	17.0	19.1	20.3	20.1	17.0	14.2	10.9	6.3	(22)
(23)		5%	DB	3.8	4.9	10.9	19.2	25.9	28.2	29.9	29.8	24.2	17.0	10.2	5.9	(23)
(24)			MCWB	2.9	3.7	6.7	11.8	16.7	18.7	20.2	19.5	16.1	13.0	9.0	5.3	(24)
(25)		10%	DB	2.1	3.1	8.2	17.1	23.8	26.2	28.0	27.2	21.9	14.8	9.0	4.0	(25)
(26)			MCWB	1.5	2.1	5.4	10.8	16.0	18.2	19.5	18.4	15.2	11.6	8.1	3.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.3	10.3	15.7	20.3	22.1	22.9	22.5	19.6	17.1	12.9	8.9	(27)
(28)			MCDWB	6.6	8.8	15.3	22.6	26.9	28.2	29.1	29.8	26.9	22.0	14.9	9.6	(28)
(29)		2%	WB	4.1	5.1	8.6	13.8	18.8	20.9	21.9	21.2	18.0	15.1	11.0	6.4	(29)
(30)			MCDWB	4.8	6.2	11.9	19.6	24.9	26.8	28.4	28.3	24.3	18.7	12.4	6.9	(30)
(31)		5%	WB	2.7	3.7	7.0	12.7	17.7	20.0	21.1	20.3	16.8	13.4	9.4	5.0	(31)
(32)			MCDWB	3.3	4.7	9.7	17.6	23.8	26.1	27.3	27.2	22.2	16.6	10.2	5.6	(32)
(33)		10%	WB	1.5	2.2	5.6	11.5	16.6	19.1	20.3	19.4	15.7	11.9	7.9	3.2	(33)
(34)			MCDWB	1.9	2.9	8.1	16.2	22.2	24.9	26.1	25.7	20.6	14.5	8.7	3.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	6.2	8.2	11.1	12.1	11.9	12.0	12.6	11.3	9.6	5.5	4.9	(35)
(36)			MCDBR	4.9	7.0	12.3	14.4	14.8	14.7	14.7	16.5	15.9	13.6	7.7	5.7	(36)
(37)			MCWBR	4.3	5.4	7.6	7.3	6.6	6.2	5.6	6.3	7.3	8.3	6.0	5.0	(37)
(38)		5% WB	MCDBR	4.6	6.4	10.9	12.3	12.8	12.6	12.6	13.6	13.3	12.1	6.9	5.6	(38)
(39)			MCWBR	4.2	5.2	7.5	6.9	6.4	6.1	5.5	5.9	7.0	7.9	5.7	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.339	0.371	0.436	0.409	0.401	0.430	0.432	0.399	0.360	0.341	0.311	(40)	
(41)		taud	2.216	2.163	2.221	2.121	2.280	2.323	2.252	2.232	2.288	2.376	2.315	2.262	(41)	
(42)		Ebn at Noon	701	773	822	803	849	858	824	800	785	753	663	641	(42)	
(43)		Edh at Noon	74	101	114	142	126	122	129	126	107	82	68	62	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.87	1.69	2.98	4.34	5.68	6.00	5.83	5.08	3.46	1.99	0.79	0.59	(44)	
(45)		RadStd	0.15	0.29	0.47	0.44	0.45	0.44	0.38	0.31	0.52	0.32	0.12	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.62	N/A	N/A	+0.98	+0.50	+0.60	N/A	-107	+111	+71
(47) Regional (10 neighbors)		+0.73	N/A	N/A	+1.00	+0.48	N/A	-140	-196	+119	+55

Nomenclature: See separate page